DynaMMo: Mining and Summarization of Co-evolving Sequences with Missing Values

Reproduction, Ablation, and Failure-Mode Analysis

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Mid-Semester Examination — Part B
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Based on: Li, McCann, Pollard, Faloutsos — KDD 2009

***Abstract*—DynaMMo [1] models d co-evolving time series through a k-dimensional Linear Dynamical System (LDS), inferring missing values by projecting EM-learned latent states through an emission matrix. This report synthesises findings from a full reproduction on a synthetic air-quality dataset (T=300, d=8, k=2, 16% missing): the reproduced model achieves RMSE=0.2249, a 68% improvement over column-mean fill. Ablations confirm both the RTS Smoother and the diagonal noise constraint are genuinely beneficial. A failure-mode sweep shows DynaMMo's advantage collapses at structured missing rates above 30%, where correlated sensor dropout violates the MCAR assumption.**

***Index Terms*—time-series imputation, Linear Dynamical System, EM algorithm, Kalman Filter, RTS Smoother, missing data.**

I. Paper Summary

DynaMMo [1] addresses missing-value imputation in multivariate co-evolving time series by exploiting their shared low-rank structure. The core insight is that d sensor sequences are driven by $k \ll d$ hidden factors, captured by a Linear Dynamical System with state transition $\mathbf{z}_t = \mathbf{A}\,\mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t$ and observation equation $\mathbf{x}_t = \mathbf{C}\,\mathbf{z}_t + \boldsymbol{\delta}_t$, where $\mathbf{A} \in \mathbb{R}^{k\times k}$ is the transition matrix, $\mathbf{C} \in \mathbb{R}^{d\times k}$ is the emission matrix, and both noise terms are Gaussian. Parameters are learned via Expectation-Maximisation: the E-step runs a Kalman Filter forward pass and an RTS Smoother backward pass [2] over the *observed* sensor dimensions at each timestep, computing posterior latent-state distributions without any pre-imputation; the M-step then updates $\mathbf{A}$, $\mathbf{C}$, and $\mathbf{R}$ in closed form from the resulting sufficient statistics. Once trained, any missing entry $\hat{x}_{t,j}$ is recovered as $[\mathbf{C}\,\boldsymbol{\mu}_t^s]_j$, the j-th component of the projected smoothed latent state. On real datasets such as UCI Chlorine (507 sensors) and CMU Motion Capture (62 joints), DynaMMo reduces RMSE by 20–40% relative to mean-fill and linear interpolation, particularly on block gaps where naive baselines have no temporal context to exploit.

II. Reproduction Setup and Results

A. Experimental Setup

The reproduction uses a synthetic LDS dataset (seed = 42) with T=300 timesteps, d=8 sensors (CO, NOx, NO_2, O_3, Temp×4), and k=2 true latent factors: a periodic traffic cycle $Z_{:,0} = \sin(t) + 0.3\sin(3t)$ and a slow AR weather drift $Z_{:,1}$. Missing values are introduced at 15% of positions via random dropout plus four structured block gaps of 5–20 steps, totalling 384 NaN entries (16.0%).

TABLE I
Overall RMSE on 384 Missing Positions vs. $\mathbf{X}_{\text{TRUE}}$

Method	RMSE	vs. DynaMMo
DynaMMo (EM)	**0.2249**	—
Column-Mean Fill	0.7090	+215%
Linear Interpolation	0.2760	+23%

The training/evaluation split is absolute: $\mathbf{X}_{\text{miss}}$ (with NaN) is the *sole* input to `model.fit()`; $\mathbf{X}_{\text{true}}$ is sealed and used exclusively to compute RMSE after training. DynaMMo is trained for 30 EM iterations with `n_dim_state=2`. Early stopping selects the iteration with the lowest validation RMSE on 20% of held-out NaN positions (76 of 384), whose ground-truth values are extracted from $\mathbf{X}_{\text{true}}$ before training and never fed into any Kalman update or M-step. $\mathbf{Q}$ is fixed at the identity throughout to prevent the identifiability collapse in which $\mathbf{A} \to \mathbf{I}$ and $\mathbf{C} \to \mathbf{0}$ when $\mathbf{Q}$ is updated freely [3].

B. Quantitative Results

DynaMMo achieves RMSE = 0.2249 and is best on all 8 sensors individually (Table I). The 68% relative improvement over mean-fill demonstrates the value of temporal and cross-sequence modelling. Early stopping triggered at iteration 8 (val-RMSE = 0.2275), after which the emission matrix continued shrinking.

C. Honest Commentary on the Gap

The paper reports 20–40% improvement over mean-fill on UCI Chlorine and CMU Motion Capture; our reproduction shows 68%. Three factors explain the larger gap: (i) our data is generated by an exact LDS so the model class is

TABLE II
ABLATION RESULTS

Configuration	RMSE	Δ	$\Delta\%$
Ablation 1: RTS Smoother			
Full DynaMMo (KF+RTS)	0.2526	—	—
No RTS Smoother	0.2619	+0.009	$+3.7\%$
Ablation 2: Diagonal $\mathbf{R}$			
Full DynaMMo (diag. $\mathbf{R}$)	0.2526	—	—
Full-covariance $\mathbf{R}$	0.2566	+0.004	$+1.6\%$

TABLE III
RMSE SWEEP UNDER STRUCTURED CORRELATED DROPOUT

Rate	DynaMMo	Mean	Linear	Best?
10%	0.247	0.732	0.295	✓ (+197%)
20%	0.395	0.787	0.325	✓ (+99%)
30%	0.457	0.763	0.393	× (Linear wins)
50%	0.713	0.804	0.608	× (Linear wins)
70%	0.701	0.956	0.905	✓ (marginal)
80%	0.712	0.962	0.867	✓ (marginal)

perfectly specified—zero model mismatch; (ii) at T=300 the cross-sequence signal is proportionally stronger; (iii) real data has non-Gaussian noise and non-stationarities that the LDS cannot fully capture, capping its advantage on real datasets. The reproduction therefore confirms the paper's directional claims but cannot be used to predict exact performance on real data without accounting for model mismatch.

III. ABLATION FINDINGS

A. Ablation 1 — Removing the RTS Smoother

The RTS Smoother backward pass refines the Kalman-filtered state $p(\mathbf{z}_t \mid \mathbf{x}_{1:t}^{\text{obs}})$ into the smoothed state $p(\mathbf{z}_t \mid \mathbf{x}_{1:T}^{\text{obs}})$ by propagating future observations backward through the sequence. Removing it forces imputation to use only *left* context, making mid-gap positions entirely dependent on AR extrapolation from the last observed point. The $+3.7\%$ RMSE increase (Table II) confirms that bidirectional context is meaningfully beneficial: for a gap of length L, the smoother receives the observation immediately after the gap and integrates it backward, reducing uncertainty proportionally rather than allowing it to grow with L.

B. Ablation 2 — Diagonal vs. Full $\mathbf{R}$

The standard DynaMMo model constrains $\mathbf{R} = \text{diag}(r_1, \ldots, r_d)$. Relaxing this to a full $d{\times}d$ covariance adds $d(d{+}1)/2 = 36$ parameters for d=8, risking overfitting. Counterintuitively, the diagonal constraint *improves* RMSE by 1.6% over unconstrained $\mathbf{R}$. The mechanism is regularisation: by preventing $\mathbf{R}$ from absorbing cross-sensor covariance that should be explained by $\mathbf{C}$, the diagonal structure forces the latent space to be more expressive. Our learned $\mathbf{R}$ diagonal spans a $4\times$ range (0.038–0.148), confirming the model differentiates sensor noise levels. Both ablations validate the original paper's design choices: the RTS Smoother and diagonal $\mathbf{R}$ are not arbitrary simplifications but genuinely performance-improving decisions.

IV. FAILURE MODE ANALYSIS

A. Scenario: High Structured Missing Rate

DynaMMo is evaluated under increasing structured missing rates (10–80%) where sensors fail together in correlated blocks, directly violating Assumption 4 (MCAR) identified in Task 1.2. Table III shows the RMSE sweep.

B. Mechanistic Explanation

Three failure mechanisms compound at high structured missing rates.

Latent state drift. When all sensors at timestep t are missing ($\mathcal{O}_t = \emptyset$), the Kalman update is skipped entirely, leaving $\boldsymbol{\mu}_{t|t} = \mathbf{A}\boldsymbol{\mu}_{t-1|t-1}$ to drift on AR dynamics alone. For consecutive fully-missing timesteps this drift accumulates, and the smoothed state $\boldsymbol{\mu}_t^s$ departs significantly from the true $\mathbf{z}_t$.

Biased emission estimation. The M-step sufficient statistic $\mathbf{P}_{xz} = \sum_t \mathbf{x}_t^{(w)} \boldsymbol{\mu}_t^{s\top}$ is accumulated only from the $\sim$30% of non-missing timesteps at 70% missing rate. Those timesteps represent the normal operating regime, not the failure regime. The learned $\mathbf{C}$ therefore aligns the latent space with non-failure dynamics, producing systematically biased imputation inside failure blocks.

Loss of cross-sensor sharing. DynaMMo's core advantage—imputing sensor j using information from sensors $i \neq j$ at timestep t—requires $|\mathcal{O}_t| \geq 1$. Structured dropout eliminates this entirely for affected timesteps. Linear interpolation, making no structural assumptions, remains competitive by simply connecting pre-gap and post-gap observations.

V. REFLECTION

A. What Could Not Be Fully Implemented

The paper reports results on UCI Chlorine (507 sensors, 4310 timesteps) and CMU Motion Capture (62 joints). These were not reproduced for three reasons: (i) CMU requires custom AMC/ASF motion-capture file parsing; (ii) scaling the Kalman Filter to d=507 requires blocked matrix operations not in the current NumPy-only code; (iii) the paper's comparison includes TRMF and MICE baselines that were out of scope for the core EM reproduction. Using a synthetic LDS dataset is deliberate: it eliminates model mismatch as a confound, providing a clean ground-truth evaluation that real datasets cannot offer.

B. What Was Surprising

The most surprising finding was the severity and speed of the C-collapse instability when $\mathbf{Q}$ is updated freely. After just 9 iterations the emission matrix norm dropped from 1.414 to 0.345, and by iteration 30 it had reached 0.071—an order of magnitude smaller than initialisation. The log-likelihood simultaneously decreased after iteration 10, appearing to violate EM's theoretical monotonicity guarantee. Tracing this to the identifiability problem ($\mathbf{A} \to \mathbf{I}$ routes all variance through $\mathbf{R}$,

leaving nothing for $\mathbf{C}$ to explain) was a key insight glossed over in the paper, and fixing $\mathbf{Q}{=}\mathbf{I}$ stabilised training immediately.

A second surprise was the non-monotonic failure-mode curve: DynaMMo loses to linear interpolation in the 30–60% structured missing range, but recovers a marginal advantage at 70–80%. The recovery occurs because at very high missing rates, linear interpolation itself degrades sharply (long gaps have no nearby anchor points), while DynaMMo's AR dynamics at least provide a bounded prediction.

C. What Would Be Revisited With More Time

Three directions would be priorities. First, **scaling to real datasets** by implementing blocked Kalman updates $O(T{\cdot}k{\cdot}d)$ using sparse matrix representations, enabling evaluation on UCI Chlorine and CMU Motion Capture to close the gap with the paper's reported numbers. Second, **learning $\mathbf{Q}$ with regularisation**—rather than fixing $\mathbf{Q}{=}\mathbf{I}$, exploring a prior-regularised M-step (e.g., $\mathbf{Q} \approx \lambda\mathbf{I}$ with λ learned via cross-validation) that prevents C-collapse while still adapting the state noise scale to the data. Third, **addressing the structured-dropout failure** by incorporating an explicit missingness indicator $\mathbf{o}_t \in \{0,1\}^d$ as an observed input to the state transition, allowing the model to learn separate AR dynamics for observed and missing regimes and eliminating the M-step bias that degrades performance at high structured missing rates.

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